Gayathri Ragupathi

Wissenschaftliche Hilfskraft (WHK)
M.Sc. Student in Electronics Design Technology | Embedded Systems
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PROFESSIONAL SUMMARY

Wissenschaftliche Hilfskraft (WHK) with hands-on experience across embedded Linux, Jetson-based edge AI deployment, FPGA-Zynq systems, ESP32, and STM32F446RE. Currently pursuing an M.Sc. in Electronics Design Technology with a focus on embedded systems, hardware/software integration, ML deployment, and technical documentation. Passionate about learning industrial embedded architecture and projects.

PROFESSIONAL EXPERIENCE

May 2026 - Present

Wissenschaftliche Hilfskraft (WHK) — Embedded AI / Edge ML, Germany
Universität Siegen

- Developed and deployed embedded AI models on NVIDIA Jetson edge hardware, including an audio-based ML pipeline for industrial pipeline leak detection and a YOLO-based carton detection model for vision-based industrial inspection.
- Configured and validated the Jetson platform by handling embedded Linux setup, hardware validation, connectivity troubleshooting, and deployment environment preparation.
- Worked on audio dataset preparation, Edge Impulse model development support, and live inference validation, strengthening hands-on skills in audio ML, edge deployment, and embedded Linux debugging.

Aug 2023 - Mar 2025

Infrastructure Associate, India
Mastech InfoTrellis

- Supported Linux infrastructure operations across Red Hat, CentOS, Rocky Linux, Ubuntu, and Debian environments, including monitoring, log analysis, networking, firewall/SELinux troubleshooting, and configuration validation.
- Worked with Docker, GitLab CI/CD, Bash scripting, VMware, and Linux deployment workflows, contributing to reliable infrastructure delivery and client readiness.

Mar 2023 - Aug 2023

Junior Research Intern, India
Renault Nissan Technology & Business Centre India

- Researched wireless EV charging systems, focusing on power transfer mechanisms, system architecture, vehicle-level integration, and electromagnetic coupling.
- Analyzed trade-offs in efficiency, control strategy, and reliability while preparing structured engineering documentation on automotive electronics, EV power systems, and embedded control concepts.

Sep 2022 - Feb 2023

ZF WABCO, India
Project Trainee

- Gained exposure to EV inverter systems, including power electronics, DC-AC conversion, motor control requirements, and vehicle-level integration.
- Supported engineering documentation and analysis for EV inverter/powertrain system concepts.

EDUCATION

Apr 2025 - Present

Universität Siegen, Germany
Masters in Electronics Design Technology
Specialization: Embedded Systems, **Grade: 2.0**

Aug 2019 - Mar 2023

Rajalakshmi Engineering College (Anna University), India
Bachelors in Electrical and Electronics Engineering, **Grade: 1.3**

SKILLS

Embedded Firmware & MCUs	PIC24 | ESP32 | Sensor Interfacing | Bangle.js | STM32F446RE |STM32Cube IDE | STM32 HAL | Embedded C | UART | Serial Debugging | Timers/Interrupts
Embedded & Edge AI	Embedded Linux | NVIDIA Jetson Orin NX/Nano | JetPack | Board Bring-Up | Hardware Debugging | Edge Impulse | Audio ML | ONNX | TensorRT | YOLO11n
FPGA & Digital Design	Zynq-7000 SoC | Xilinx Vivado | Vitis | VHDL | HW/SW Co-Design | EEPROM | Python | C | C++ | Bash | MATLAB
Deployment	Docker | GitLab CI/CD | VMware Workstation | Linux OS

PROJECTS AND ACHIEVEMENTS

- **STM32 Industrial Sensor Node - Embedded C Firmware Project | 2026**
  STM32F446xx · STM32CubeIDE · Embedded C · STM32 HAL · ADC · UART · GPIO · Git/GitHub
  - Completed an STM32-based sensor-node firmware project on STM32F446RE, configuring GPIO, ADC, UART, and timer-driven sampling in STM32CubeIDE using Embedded C and STM32 HAL.
  - Implemented ADC raw-value acquisition, millivolt/voltage conversion, temperature conversion and calibration logic, formatted UART debugging, and serial-monitor validation for real-time sensor output.
  - Gained practical experience in peripheral configuration, embedded C debugging, hardware/software validation, threshold alert logic, moving-average filtering, Git/GitHub documentation, and firmware bring-up workflows.

- **Embedded Edge AI Deployment for Industrial Audio and Vision Inspection | 2026 WHK**
  Embedded Linux · Edge AI · Audio ML · Computer Vision · NVIDIA Jetson Orin NX/Nano · JetPack · Edge Impulse · YOLO11n · Python
  - Set up and configured NVIDIA Jetson Orin NX/Nano platforms for edge AI deployment, including JetPack installation and basic hardware validation. Developed an audio-based ML model for pipeline leak detection (1 m, 3 m, no-leak) using UMA16 data and Edge Impulse, deployed on AARCH64 Linux.
  - Built a YOLO11n-based carton detection model on Jetson Nano, converting annotations and validating on-device inference. Prepared and organized an object-detection dataset with 385 images and 1,459 bounding boxes using an 80/20 train-validation split.

- **FPGA Embedded Systems — Zynq-7000 SoC Hardware/Software Co-Design | 2025 - 2026**
  FPGA · Zynq XC7Z020 · Vivado · Vitis · VHDL · HW/SW Co-Design
  - Designed FPGA-based embedded systems on Zynq-7000 XC7Z020 using VHDL, Vivado synthesis/implementation, bitstream generation, and Vitis hardware handoff.
  - Integrated programmable logic with the ARM Cortex-A9 processor system, gaining practical experience in HW/SW co-design, custom IP workflows, and FPGA-to-processor communication.

- **Embedded Firmware & IoT Projects | 2025**
  PIC24 · ESP32 · Bangle.js · EEPROM · Blynk IoT · Espruino · Wearable Embedded
  - Developed embedded applications across PIC24, ESP32, and Bangle.js platforms, including a door access system, smart energy monitor, and wearable emotion display.
  - Implemented EEPROM-based user storage, Wi-Fi sensor monitoring, OLED/Blynk data visualization, and wearable UI logic on resource-constrained embedded devices.

LANGUAGES

English (Fluent) | German / Deutsch A2 (actively improving) | Tamil (Native)

HOBBIES

Dance (Free style) | Cooking | Sports (Badminton)

Gayathri Ragupathi | July 2026